

Controlling Experience for Interaction Techniques in Virtual Reality Exergames

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ABSTRACT

Virtual reality (VR) has become an essential platform for upper limb rehabilitation by introducing VR interaction techniques (e.g., hand gestures, controllers, and tangible tools) into traditional rehabilitation apparatus. Controlling experience plays a significant role in upper limb exergames, which is highly related to rehabilitation outcomes. This study focused on users' controlling experience when integrating VR interaction techniques and traditional rehabilitation tables (commonly employed in clinics and homes). This study revealed that tangible tools or controllers contributed to the higher controlling experience, and they were recommended as the suitable options for upper limb exergames when building VR rehabilitation.

Keywords: Virtual reality rehabilitation, upper limb exergame, controlling experience, rehabilitation table.

Index Terms: H.1.2 [Human-centered computing]: Human computer interaction (HCI) — HCI design and evaluation methods — User studies

1 INTRODUCTION

Virtual reality (VR) is emerging as a promising platform for upper limb rehabilitation, offering transformative benefits to patients with motor deficits [1]. VR enables traditional rehabilitation tasks into enjoyable experiences, boosting user motivation and engagement [1]. When introducing VR into upper limb rehabilitation, it is essential to consider the optimal interaction techniques for users to engage in virtual settings. Related work has suggested using traditional VR interaction techniques, such as hand gestures [2] and controllers [3], for upper limb rehabilitation exergames. In traditional rehabilitation settings, many upper limb exercises practically depend on specialized apparatus, such as rehabilitation tables, which are widely used in clinics and homes throughout China [4]. Hence, integrating specialized apparatus from traditional rehabilitation settings with interaction techniques in VR poses a new and key challenge. However, little has been known in the existing literature regarding the optimal interaction techniques in VR rehabilitation when integrated with physical rehabilitation apparatus.

This study explores interaction techniques in VR integrated with rehabilitation apparatus, specifically the widely employed rehabilitation tables in clinical and home settings. An interactive rehabilitation system is developed to combine a rehabilitation table with VR interaction, providing upper limb exergames in correspondence to traditional rehabilitation scenarios. When considering the interaction technique in VR exergames, it is crucial to focus on users' controlling experience, which enables them to engage in exergames effectively [1, 2]. Controlling experience, derived from the concept of the sense of agency in neuroscience, refers to feeling in control over one's actions [5, 6]. Scholars in human-computer interaction proposed the controlling experience as a key user experience factor and established design guidelines for interactive systems to enhance the controlling experience [1]. In the context of rehabilitation, patients with upper limb motor deficits often experience diminished controlling experience compared with healthy people, so enhancing

controlling experience is a crucial aspect of their rehabilitation routines [1].

Therefore, this study concentrates on integrating VR interaction techniques with traditional rehabilitation tables and explores the influence of VR interaction techniques on users' controlling experience for upper limb exergames. The contributions of this study are listed as follows: (1) generating new insights into users' controlling experience with different VR interaction techniques in upper limb exergames; (2) offering a new and practical reference for designing VR interaction techniques for exergames with traditional rehabilitation apparatus integrated.

2 USER STUDY

Twelve participants (six males and six females) were enrolled in the user study with an average age of 21 ± 3 years. Prior to the experiment, they received information about this study and signed a consent form. This study has received ethical approval from the University Ethics Committee.

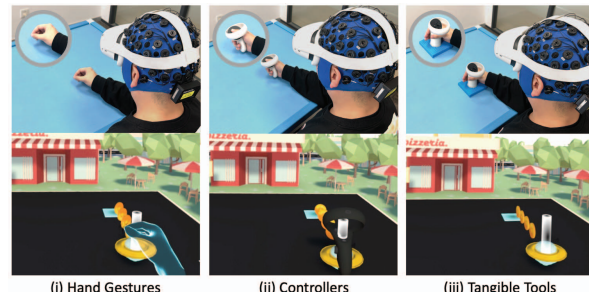


Figure 1: Three interaction techniques in the experiment.

In this study, the VR exergame was designed and implemented in a scene of a user moving a small vehicle inside a virtual town. Based on the VR exergame, participants wearing a VR headset (Oculus Rift) were required to complete some experimental tasks by moving the virtual vehicle with different interaction techniques (hand gestures, controllers, and tangible tools).

- *Hand gestures.* Participants employed the hand grasping gesture (Figure 1(i)) to grasp the virtual vehicle to perform the moving tasks.
- *Controllers.* Participants adopted an Oculus Rift controller (Figure 1(ii)) by pressing its trigger to perform tasks of moving the vehicle.
- *Tangible tools.* A tangible tool, different from the other two interactions as stated above, was designed (Figure 1(iii)) based on Lego bricks, a 3D printed holder, and a controller (for tracking the tangible tool), simulating the traditional scenario on the rehabilitation table. Participants held the tangible tool to conduct the moving tasks.

The vehicle routes are defined based on four kinds of upper limb exercise routines displayed in Figure 2, which are common forms of upper limb exercises on the rehabilitation table in traditional clinics. Specifically, Figure 2(i)(ii) shows the shoulder and elbow joints flexion-extension for left and right upper limbs; meanwhile, Figure 2(iii)(iv) illustrates the shoulder rotation and elbow flexion-extension for left and right upper limbs. According to the four routines, participants were required to complete four

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experimental tasks in VR while receiving physical support from a real rehabilitation table. Three interaction techniques were used by participants in a random order to conduct experimental tasks.

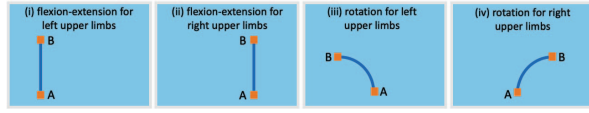


Figure 2: Upper limb exergames on rehabilitation table in VR.

This user study assessed participants' controlling experience when conducting exergames with various interaction techniques through subjective and objective measures, as listed below.

- **Self-report measure.** A commonly-used questionnaire consisting of four items was utilized to evaluate participants' controlling experience [6].
- **Performance measure.** The exergame program automatically collected the task completion time, providing an alternative perspective on the assessment of controlling experience.
- **Electroencephalogram (EEG) measure.** Alpha power analysis was used to explore the controlling experience based on the EEG data recorded by an EEG device (Neuracle NeuSen W with saline sensors) worn on the participant's head, as presented in Figure 2.

3 RESULTS AND DISCUSSION

As displayed in Figure 3(i), participants rated the lowest control scores to hand gestures ($M=5.83$), and they assigned controllers ($M=6.44$) with lower control scores compared to tangible tools ($M=6.71$). Moreover, the Kruskal-Wallis test revealed significant differences in control scores between hand gestures and controllers ($\chi^2=5.14, p=0.04$), as well as between hand gestures and tangible tools ($\chi^2=6.24, p=0.01$).

From the subjective perspective, participants reported higher controlling experience while performing upper limb exergames with tangible tools compared to the other two interactions. These findings may be attributed to users feeling more engaged when holding tangible tools in their hands. In addition, tangible tools offered more support from the rehabilitation table, allowing users to better complete VR exergames on the table. Simultaneously, tangible tools offered flexibility, enabling users to extend various reusable tools with Lego bricks.

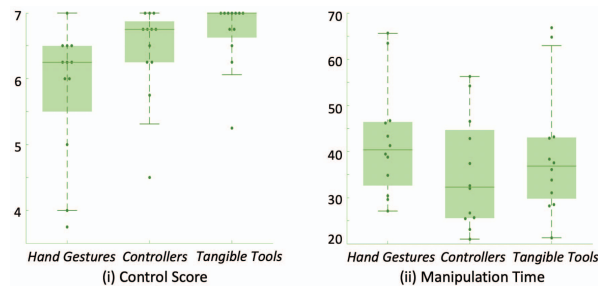


Figure 3: The boxplot of (i) control score and (ii) manipulation time.

Figure 3(ii) shows no significant differences in manipulation time among three distinct interaction techniques via the Kruskal-Wallis test. Participants spent more time applying hand gestures ($M=42.23$) to perform these exergames than controllers ($M=35.31$) and tangible tools ($M=39.40$).

Prior work showed that users felt higher controlling experience when performing efficiently [2, 3]; hence, based on performance measure, participants felt higher controlling experience with less manipulation time when using controllers for VR exergames. The outcomes could be attributed to the fact that controllers offered

participants tangible objects and delivered decreased friction from the rehabilitation table.

EEG alpha power was employed to evaluate users' controlling experience. The Wilcoxon signed-rank test revealed significant differences in alpha power at the frontal brain regions among interaction techniques, as presented in Figure 4. Users reported the highest frontal alpha power when employing hand gestures across interaction techniques in exergames, while the other two presented a comparable level of frontal alpha power.

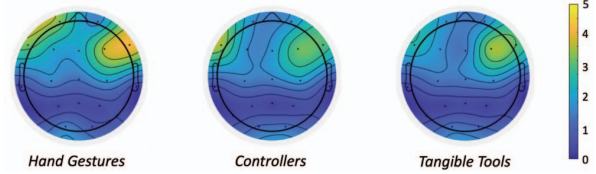


Figure 4: The brain topographic map of alpha power.

The findings from EEG data showed that tangible tools and controllers resulted in lower frontal alpha power than hand gestures. Previous work found a negative relationship between frontal alpha power and changes in controlling experience [5, 6]. Moreover, frontal lobes are related to brain networks that enable motor control [5]. Hence, this study revealed that users perceived increased controlling experience while employing controllers and tangible tools in VR exergames.

4 CONCLUSION

This study focused on the integration of VR interaction techniques and traditional rehabilitation tables for upper limb exergames. The results presented that tangible tools and controllers led to higher controlling experience, which is a key factor in VR rehabilitation. Since tangible tools have the potential to extend more interaction possibilities, this study recommends tangible tools as the suitable options for VR exergames combined with rehabilitation tables. Further work would be undertaken in a local hospital to assess patients' controlling experience for upper limb exergames in the VR rehabilitation system.

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